Bruce Guo

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Experience

Electronic Arts May 2025 – Aug. 2026
Graduate Research Intern (Mitacs) *Edmonton, AB*

- Reverse-engineered the Unreal Engine Blueprint text format through black-box analysis and implemented text generation from SQL records for a Blueprint diff tool.
- Developed Go query-generation tools for an internal RAG agent used by 315 users, translating LLM-extracted intent into valid Elasticsearch aggregation queries for summaries and reports.
- Analyzed user conversations to identify agent failures and debugged issues in internal data retrieval and analysis workflows.

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, SQL
Frameworks: React, Next.js, FastAPI, Celery, Tailwind CSS
Databases: PostgreSQL, Redis, Elasticsearch
Tools: Docker, Docker Compose, Nginx, Linux, Supabase, pytest

Projects

Oak | *Next.js, TypeScript, PostgreSQL, Supabase, Gemini API, Docker* chatwithoak.com

- Designed and built a full-stack LLM chat app with persistent branching conversations, allowing users to open side threads from selected text and merge them into the main conversation.
- Implemented Next.js APIs for streaming Gemini responses through server-sent events (SSE) and optional context sharing between main and branch conversations.
- Modeled conversations, messages, and branch relationships in PostgreSQL, supporting persistent histories, ordered messages, and cascading deletion.
- Integrated Google sign-in through Supabase Auth and deployed the app on a Linux server using Docker and Nginx.

Atex | *Next.js, TypeScript, FastAPI, Python, Celery, Redis, Supabase, Docker* atex.ink

- Built a full-stack LaTeX editor with multi-file editing, autosave, and side-by-side PDF previews using React and FastAPI.
- Moved synchronous compilation to Redis-backed Celery workers, offloading multi-file Tectonic builds from API requests and delivering generated PDFs through signed URLs.
- Integrated Gemini for academic rewrites, validating structured responses and designing prompts to preserve LaTeX commands.
- Orchestrated separate API, compilation worker, Redis, and Nginx services using Docker Compose.

Research

University of Alberta
Master's Thesis Research *Edmonton, AB*

- Built data mining pipelines for 10,000 Unreal Engine and 1,355 Unity repositories, extracting approximately 347,000 Blueprint graphs and collecting C++, C#, and Unity Visual Scripting source code.
- Developed a custom parser for text, binary, and mixed-format Unity Visual Scripting assets, including nested structures; built Python metric calculators with approximately 80 automated tests using pytest.
- Proposed and validated three adapted software metrics for visual code and compared metric magnitudes across visual and textual code.

Education

University of Alberta Edmonton, AB
M.Sc. in Computing Science (Thesis) *Sept. 2026 (requirements completed)*

Vanderbilt University Nashville, TN
B.Sc. in Computer Science *Dec. 2023*